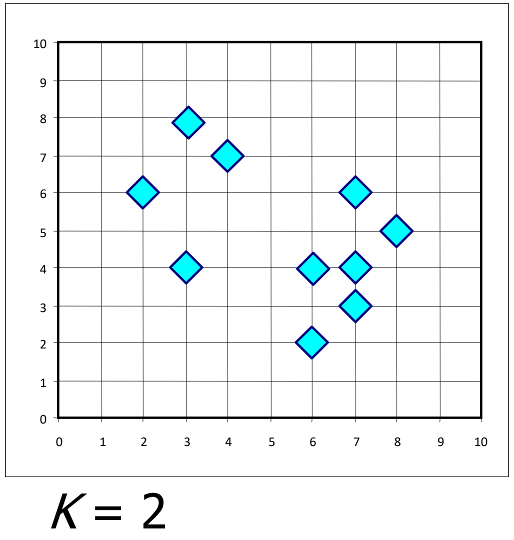
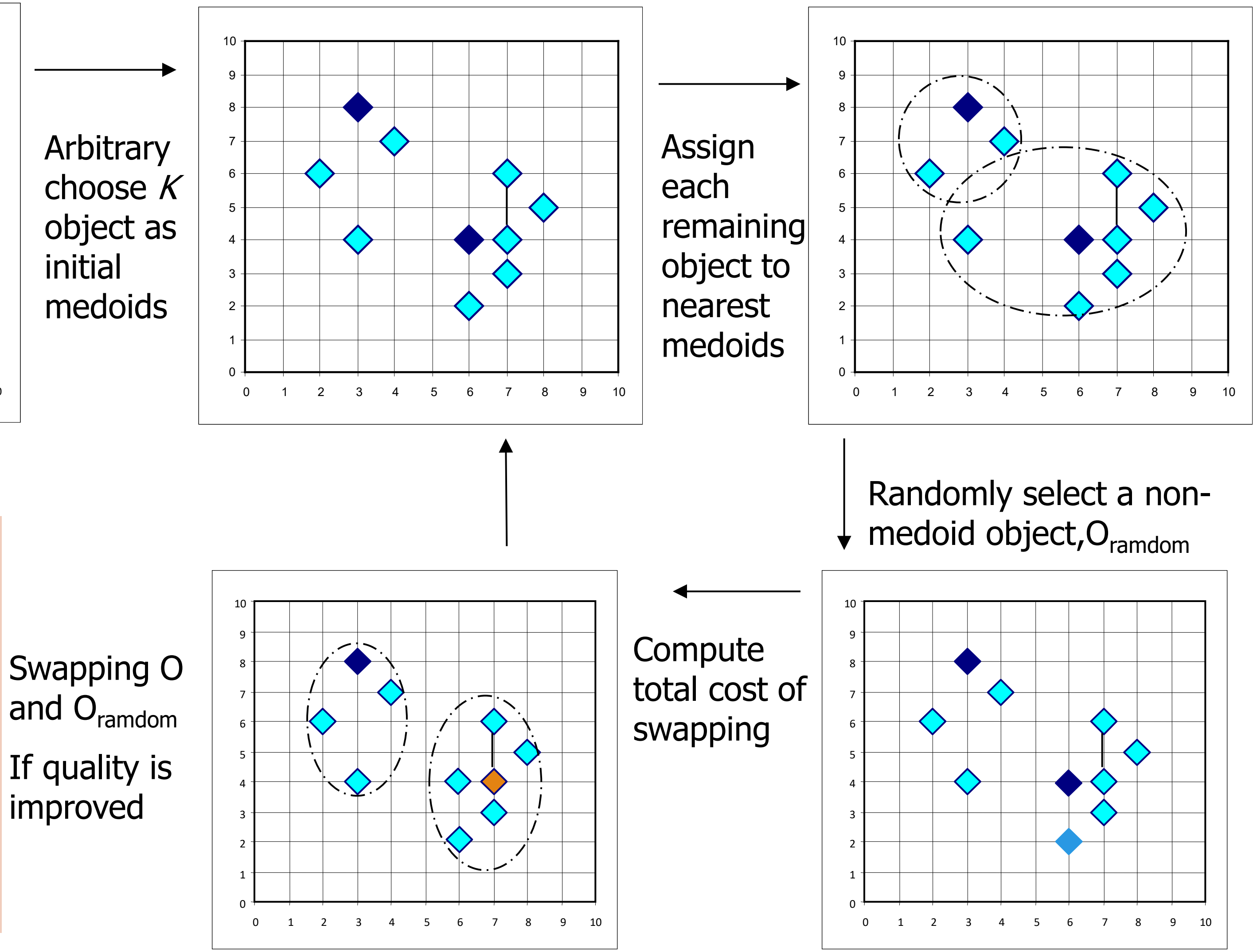
Handling Outliers: From K-Means to K-Medoids

- The K-Means algorithm is sensitive to outliers
  - An object with an extremely large value may substantially distort the distribution of the data
- K-Medoids: Instead of taking the mean value of the object in a cluster as a reference point, medoids can be used, which is the most centrally located object in a cluster
- The K-Medoids clustering algorithm:
  - Select K points as the initial representative objects (i.e., as initial K medoids)
  - Repeat
    - Assigning each point to the cluster with the closest medoid
    - Randomly select a non-representative object o_i
    - Compute the total cost S of swapping the medoid m with o_i
    - If S < 0, then swap m with o_i to form the new set of medoids
  - Until convergence criterion is satisfied

PAM: A Typical K-Medoids Algorithm





Select initial *K medoids* randomly

Repeat

Object re-assignment

Swap medoid *m* with o_i if it improves the clustering quality

Until convergence criterion is satisfied